

Pset 2

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1. $E_A = \frac{e^2}{a_0^2}$

a) Energy of the electrostatic potential:

$$V(a_0) = \frac{e^2}{a_0}$$

Energy of quantum confinement:

From the uncertainty principle, we have $\Delta x \Delta p \geq \frac{\hbar}{2}$. For $\Delta x \sim a_0$, we have $\Delta p \sim \frac{\hbar}{a_0}$.

Since the ground state wavefunction is spherically symmetric, $\langle x \rangle = \langle p \rangle = 0$. Then $\Delta p = \sqrt{\langle p^2 \rangle - \langle p \rangle^2} = \sqrt{\langle p^2 \rangle} \sim \frac{\hbar}{a_0}$, so

$\langle p^2 \rangle \sim \frac{\hbar^2}{a_0^2}$. From this, we see quantum confinement requires a kinetic energy on the order of $\frac{\langle p^2 \rangle}{2m} \sim \frac{\hbar^2}{ma_0^2}$. Setting the

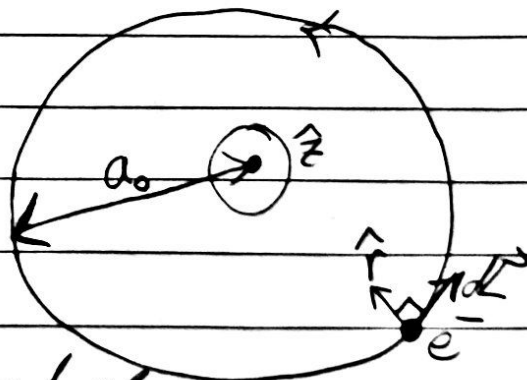
kinetic energy equal to the electrostatic potential,

$$V(a_0) = \frac{\langle p^2 \rangle}{2m} \Rightarrow \frac{e^2}{a_0} = \frac{\hbar^2}{ma_0^2} \Rightarrow a_0 = \frac{\hbar^2}{me^2}$$

b) $I = \frac{e}{\tau} = \frac{ev}{2\pi a_0}$, where v is



the speed of the electron.



Treating the electron's orbit as a current loop with current I ,

$$d\vec{B} = \frac{1}{c} \frac{I d\vec{L} \times \hat{r}}{r^2}, \quad d\vec{L} \perp \hat{r}$$

$$\Rightarrow B_N = \frac{I}{ca_0^2} \oint dL = \frac{I}{ca_0^2} 2\pi a_0 = \frac{2\pi I}{ca_0} = \frac{2\pi e v}{2\pi a_0 \cdot ca_0} = \frac{e v}{ca_0^2}.$$

In terms of $L = n\hbar$,

$$B_N = \frac{\overbrace{e m v a_0}^L}{c m a_0^3} = \frac{e \overbrace{p a_0}^L}{c m a_0^3} = \frac{e L}{c m a_0^3} = \frac{e n \hbar}{c m a_0^3}.$$

For the ground state, $n=1$, so

$$B_N = \frac{e \hbar}{c m a_0^3}.$$

If we instead find v by balancing the centripetal acceleration with the electrostatic force, $\frac{mv^2}{a_0} = \frac{e^2}{a_0^2} \Rightarrow v = \sqrt{\frac{e^2}{m a_0}} = \frac{e}{\hbar}$

$$\Rightarrow \left[B_N = \frac{e^3}{c a_0^2 \hbar} = \frac{m e^3 \hbar}{c m a_0^2 \hbar^2} = \frac{e \hbar}{c m a_0^3} \right], \text{ the same result as before.}$$

c) One Hartree is given by $1 \text{ hartree} = \frac{e^4 m}{\hbar^2}$

Assuming $\vec{\mu}$, with $|\vec{\mu}| = \mu_B$, is aligned with B_H ,

$$|\text{interaction energy}| = |B_H \mu_B| = \frac{e^4 m}{\hbar^2} \Rightarrow B_H = \frac{e^4 m}{\hbar^2 \mu_B} = \frac{e^4 m \cdot 2mc}{\hbar^2 e \hbar} \\ \left(\mu_B = \frac{e \hbar}{2mc} \text{ in Gaussian units} \right) = \frac{2e^3 m^2}{\hbar^3} = \frac{2emc}{\hbar a_0}$$

Ignoring factors of 2,

$$B_H = \frac{emc}{\hbar a_0}.$$

$$d) \quad B_N = \frac{\hbar}{cma_0} E_A = \frac{e^2}{\hbar c} E_A = \alpha E_A$$
$$B_H = \frac{ma_0 c}{\hbar} E_A = \frac{\hbar c}{e^2} E_A = \frac{1}{\alpha} E_A$$

e) We can calculate the size of the fields in T.

$$B_N \approx \frac{1}{137} \cdot \frac{4.803 \times 10^{-10} \text{ Fr}}{\underbrace{(5.29 \times 10^{-9} \text{ cm})^2}_{E_A}} \approx 1.25 \times 10^5 \text{ G} = 12.5 \text{ T}$$

$$B_H \approx 137 \cdot E_A \approx 2.35 \times 10^9 \text{ G} = 2.35 \times 10^5 \text{ T}.$$

Since B_H is much larger than any magnetic field that could be measured in an experiment, B_N would be preferred over B_H .

2. a) In terms of α , the Hydrogen energy spectrum is given by

$$E_n = -\frac{1}{2} \alpha^2 m_e c^2 \cdot \frac{1}{n^2}, \quad \text{where } \alpha = \frac{e^2}{\hbar c}.$$

$$\Rightarrow \hbar \omega_{m \rightarrow n} = \frac{1}{2} \alpha^2 m_e c^2 \left(\frac{1}{m^2} - \frac{1}{n^2} \right)$$

$$\Rightarrow \frac{2\pi c}{\lambda} = \frac{1}{2} \frac{\alpha^2 m_e c^2}{\hbar} \left(\frac{1}{m^2} - \frac{1}{n^2} \right)$$

$$\Rightarrow \frac{1}{\lambda} = \frac{\alpha^2 m_e c}{4\pi \hbar} \left(\frac{1}{m^2} - \frac{1}{n^2} \right)$$

$$\Rightarrow R_\infty = \frac{\alpha^2 m_e c}{4\pi \hbar} \Rightarrow f_\infty = \frac{\alpha^2 m_e c^2}{4\pi \hbar} = \frac{\alpha^2 m_e c^2}{2h}$$

Solving for α , $\alpha = \sqrt{\frac{2h f_\infty}{m_e c^2}}$, or if

we define $f_e = \frac{m_e c^2}{h}$, $\alpha = \sqrt{\frac{2f_\infty}{f_e}}$.

Since c is known exactly in SI units,



Due to the way the meter is defined, the result $\alpha = \sqrt{2 \cdot \frac{h}{m_e} \cdot \frac{R_{\infty}}{c}}$ only depends on experimental values of R_{∞} and $\frac{h}{m_e}$.

b) The de Broglie wavelength is given by

$$\lambda = \frac{h}{p} = \frac{h}{mv}, \text{ so}$$

$$\boxed{\frac{h}{m} = \lambda v}. \Rightarrow \frac{h}{m} \text{ can be obtained from measurements of } \lambda \text{ and } v \text{ of the neutron beam.}$$

c) The recoil velocity can be found from conservation of momentum:

$$p_{\text{atom}, i} + p_{\text{photon}} = p_{\text{atom}, f} \Rightarrow mv_i + \frac{h\nu}{c} = mv_f$$

$$\Rightarrow \Delta v = v_f - v_i = \frac{h\nu}{mc}$$

$\Rightarrow$ The atom's velocity changes by an amount $\boxed{v_R(v) = \frac{h\nu}{mc}}$.

$$\Rightarrow \boxed{\frac{h}{m} = \frac{c v_R(v)}{\nu}}, \text{ so } \frac{h}{m} \text{ can be obtained by}$$

measuring the recoil velocity of an atom after emitting a photon of frequency ν .

d) In this case,

$$p_{\text{atom}, i} + \underbrace{p_{\text{photon}, \nu_1}}_{\text{absorbed photon}} = p_{\text{atom}, f} + \underbrace{p_{\text{photon}, \nu_2}}_{\text{emitted photon}}$$

$$\Rightarrow m v_i + \frac{h \nu_1}{c} = m v_f + \frac{h \nu_2}{c}$$

$$\Rightarrow \Delta v = v_f - v_i = \frac{h}{m c} (\nu_1 - \nu_2)$$

$$\Rightarrow \boxed{\nu_R(\nu_1, \nu_2) = \frac{h(\nu_1 - \nu_2)}{m c}} \Rightarrow \frac{h}{m_{\text{atom}}} = \frac{c \nu_R(\nu_1, \nu_2)}{(\nu_1 - \nu_2)}$$

From conservation of energy,

$$\frac{1}{2} m v_i^2 + h \nu_1 = \frac{1}{2} m v_f^2 + h \nu_2$$

Since the atomic beam is well collimated, the atom is initially at rest in the horizontal direction, so $v_i = 0$ and $v_f = \nu_R(\nu_1, \nu_2)$.

$$\Rightarrow h \nu_1 = \frac{1}{2} m [\nu_R(\nu_1, \nu_2)]^2 + h \nu_2$$

$$\Rightarrow [\nu_R(\nu_1, \nu_2)]^2 = \frac{2h}{m} (\nu_1 - \nu_2) \Rightarrow \boxed{|\nu_R(\nu_1, \nu_2)| = \sqrt{\frac{2h}{m} (\nu_1 - \nu_2)}}$$



Setting the two expressions for $\nu_R(\nu_1, \nu_2)$ equal,

$$\Rightarrow \left| \frac{h(\nu_1 - \nu_2)}{mc} \right| = \sqrt{\frac{2h}{m}(\nu_1 - \nu_2)}$$

$$\Rightarrow \frac{h^2}{m^2 c^2} (\nu_1 - \nu_2)^2 = \frac{2h}{m} |\nu_1 - \nu_2|$$

$$\Rightarrow \boxed{\frac{h}{m_{\text{atom}}} = \frac{2c^2}{|\nu_1 - \nu_2|}}, \text{ so } \frac{h}{m_{\text{atom}}} \text{ can be obtained by}$$

measuring only the Doppler shift, $|\nu_1 - \nu_2|$, of an atomic resonance due to atom recoil.

e) The energy of the gamma ray must equal the energy difference between the the nuclear energy levels:

$$\Delta m \cdot c^2 = \frac{hc}{\lambda}$$

$$\Rightarrow \boxed{\frac{h}{\Delta m} = \lambda c}, \text{ so } \frac{h}{\Delta m} \text{ can be obtained}$$

from a measurement of λ .

3. a) For $1s$, $n=1$, $l=0$, and

$$\psi(\vec{r}_1, \vec{r}_2) = \frac{1}{\sqrt{\pi}} \left(\frac{Z}{a_0} \right)^{3/2} e^{-\frac{Z}{a_0} r_1} \cdot \frac{1}{\sqrt{\pi}} \left(\frac{Z}{a_0} \right)^{3/2} e^{-\frac{Z}{a_0} r_2}, \quad Z=2$$

$$= \frac{8}{\pi a_0^3} e^{-\frac{2}{a_0}(r_1 + r_2)}$$

$$\text{Then } \left\langle \frac{e^2}{r_{12}} \right\rangle = \left(\frac{8e}{\pi a_0^3} \right)^2 \int \frac{e^{-\frac{4}{a_0}(r_1 + r_2)}}{|\vec{r}_1 - \vec{r}_2|} d^3\vec{r}_1 d^3\vec{r}_2$$

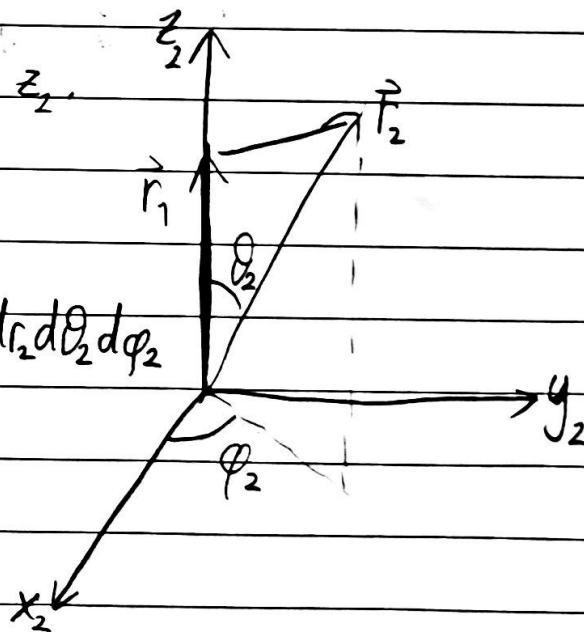
Hold $\vec{r}_1$ fixed & integrate over $\vec{r}_2$:

Define axes so $\vec{r}_1$ is along z_2 .

Then

$$|\vec{r}_1 - \vec{r}_2| = \sqrt{r_1^2 + r_2^2 - 2r_1 r_2 \cos \theta_2}$$

$$\Rightarrow \int \frac{e^{-4r_2/a_0}}{|\vec{r}_1 - \vec{r}_2|} d^3\vec{r}_2 = \int \frac{e^{-4r_2/a_0} r_2^2 \sin \theta_2 dr_2 d\theta_2 d\phi_2}{\sqrt{r_1^2 + r_2^2 - 2r_1 r_2 \cos \theta_2}}$$



The θ_2 integral is

$$\int_0^\pi \frac{\sin \theta_2 d\theta_2}{\sqrt{r_1^2 + r_2^2 - 2r_1 r_2 \cos \theta_2}} = \frac{1}{2r_1 r_2} \int_{u(0)}^{u(\pi)} u^{-1/2} du = \frac{\sqrt{r_1^2 + r_2^2 - 2r_1 r_2 \cos \theta}}{r_1 r_2} \Big|_0^\pi$$

$$\left(\begin{aligned} u &= r_1^2 + r_2^2 - 2r_1 r_2 \cos \theta_2 \\ du &= 2r_1 r_2 \sin \theta_2 d\theta_2 \end{aligned} \right) = \frac{\sqrt{r_1^2 + r_2^2 + 2r_1 r_2}}{r_1 r_2} - \frac{\sqrt{r_1^2 + r_2^2 - 2r_1 r_2}}{r_1 r_2}$$



$$\Rightarrow O \text{ integral} = \frac{\sqrt{(r_1+r_2)^2}}{r_1 r_2} - \frac{\sqrt{(r_1-r_2)^2}}{r_1 r_2}$$

$$= \frac{1}{r_1 r_2} [(r_1+r_2) - |r_1-r_2|]$$

$$= \begin{cases} 2/r_1, & r_2 < r_1 \\ 2/r_2, & r_2 > r_1 \end{cases}$$

Then $\int \frac{e^{-4r_2/a_0}}{|\vec{r}_1 - \vec{r}_2|} d^3\vec{r}_2 = 2\pi \left[\int_0^{r_1} e^{-4r_2/a_0} r_2^2 \cdot \frac{2}{r_1} dr_2 + \int_{r_1}^{\infty} e^{-4r_2/a_0} 2r_2 dr_2 \right]$

$$= 4\pi \left[\frac{1}{r_1} \frac{\partial^2}{\partial c^2} \int_0^{r_1} e^{-cr_2} dr_2 - \frac{\partial}{\partial c} \int_{r_1}^{\infty} e^{-cr_2} dr_2 \right], \quad c = \frac{4}{a_0}$$

$$= 4\pi \left[\frac{1}{r_1} \frac{\partial^2}{\partial c^2} \left(\frac{e^{-cr_2}}{-c} \right) \Big|_0^{r_1} - \frac{\partial}{\partial c} \left(\frac{e^{-cr_2}}{-c} \right) \Big|_{r_1}^{\infty} \right]$$

$$= 4\pi \left[\frac{1}{r_1} \frac{\partial}{\partial c} \left(\frac{cr_2 e^{-cr_2} + e^{-cr_2}}{c^2} \right) \Big|_0^{r_1} - \left(\frac{cr_2 e^{-cr_2} + e^{-cr_2}}{c^2} \right) \Big|_{r_1}^{\infty} \right]$$

$$= 4\pi \left[\frac{1}{r_1} \left(\frac{-r_2^2 e^{-cr_2} \cdot c - r_2 e^{-cr_2}}{c^2} + \frac{(-r_2 c^2 e^{-cr_2} - 2c e^{-cr_2})}{c^4} \right) \Big|_0^{r_1} + \frac{r_1 e^{-cr_1}}{c} + \frac{e^{-cr_1}}{c^2} \right]$$

$$= 4\pi \left[\frac{1}{r_1} e^{-cr_1} \left(-\frac{r_1^2}{c} - \frac{r_1}{c^2} - \frac{r_1}{c^2} - \frac{2}{r_1 c^3} \right) + \frac{2}{r_1 c^3} + \frac{r_1 e^{-cr_1}}{c} + \frac{e^{-cr_1}}{c^2} \right]$$

$$= 4\pi \left[e^{-cr_1} \left(-\frac{1}{c^2} - \frac{2}{r_1 c^3} \right) + \frac{2}{r_1 c^3} \right]$$

$$= 4\pi \cdot \frac{2}{r_1 c^3} \left[1 - \left(1 + \frac{r_1 c}{2} \right) e^{-cr_1} \right]$$

$$= \frac{\pi a_0^3}{8r_1} \left[1 - \left(1 + \frac{2r_1}{a_0} \right) e^{-4r_1/a_0} \right]$$

$$\Rightarrow \left\langle \frac{e^2}{r_{12}} \right\rangle = \left(\frac{8e^2}{\pi a_0^3} \right) \int \left[1 - \left(1 + \frac{2r_1}{a_0} \right) e^{-4r_1/a_0} \right] e^{-4r_1/a_0} r_1 \sin\theta_1 dr_1 d\theta_1 d\phi_1$$

$$= 4\pi \left(\frac{8e^2}{\pi a_0^3} \right) \int_0^\infty \left[r_1 e^{-4r_1/a_0} - r_1 e^{-8r_1/a_0} - \frac{2r_1^2}{a_0} e^{-8r_1/a_0} \right] dr_1$$

$$= 4\pi \left(\frac{8e^2}{\pi a_0^3} \right) \left[\left(\frac{\frac{4}{a_0} r_1 e^{-4r_1/a_0} + e^{-4r_1/a_0}}{16/a_0^2} \right) \Big|_0^\infty + \left(\frac{\frac{8}{a_0} r_1 e^{-8r_1/a_0} + e^{-8r_1/a_0}}{64/a_0^2} \right) \Big|_0^\infty \right. \\ \left. - \frac{2}{a_0} \left(\frac{-r_1^2 e^{-8r_1/a_0} \cdot \frac{8}{a_0} - r_1 e^{-8r_1/a_0}}{64/a_0^2} + \frac{(-r_1 \frac{64}{a_0^2} e^{-8r_1/a_0} \frac{16}{a_0} e^{-8r_1/a_0})}{8^4/a_0^4} \right) \Big|_0^\infty \right]$$

$$= 4\pi \left(\frac{8e^2}{\pi a_0^3} \right) \left[\frac{a_0^2}{16} - \frac{a_0^2}{64} - \frac{2}{a_0} \left(\frac{216}{a_0} \cdot \frac{a_0^2}{8^4} \right) \right]$$

$\frac{a_0^2}{64 \cdot 2}$



$$= 4\pi \left(\frac{8e^2}{\pi a_0^3} \right) a_0^2 \left[\frac{8-2-1}{128} \right]$$

$$= \frac{e^2}{a_0} \cdot \frac{5}{4}$$

$$\Rightarrow \boxed{\left\langle \frac{e^2}{r_{12}} \right\rangle = \frac{5e^2}{4a_0}} \quad \begin{array}{l} \downarrow \\ = -\frac{5}{2} E_1 \end{array} \quad \begin{array}{l} \left(E_n = -\frac{1}{2} \frac{e^2}{a_0} \frac{1}{n^2} \right) \\ \uparrow \\ \text{of Hydrogen.} \end{array}$$

$$\Rightarrow \boxed{\left\langle \frac{e^2}{r_{12}} \right\rangle = 34 \text{ eV}}.$$

Then the ground state energy is

$$\boxed{E_{gs} = -108.848 \text{ eV} + 34 \text{ eV} = -74.848 \text{ eV}}$$

b) For this part, we use the 1s wavefunction again, but with Z replaced with Z' .

The integral for $\left\langle \frac{e^2}{r_{12}} \right\rangle$ is the same, just with

$$\frac{Z}{a_0} \rightarrow \frac{Z'}{a_0}.$$

So we can get the result immediately from the result of part (a) by replacing a_0 in the expression with $2a_0/Z'$:

$$\left\langle \frac{e^2}{r_{12}} \right\rangle = \frac{5e^2}{4\left(\frac{2a_0}{Z'}\right)} = \frac{5}{8} \frac{e^2 Z'}{a_0} = -\frac{5}{4} Z' E_1$$

Then the expectation value of the ground state energy is

$$\langle H \rangle = \left\langle -\frac{\hbar^2}{2m} (\nabla_1^2 + \nabla_2^2) - e^2 \left(\frac{2}{r_1} + \frac{2}{r_2} - \frac{1}{r_{12}} \right) \right\rangle$$

Helium
Hamiltonian

$$= \left\langle -\frac{\hbar^2}{2m} (\nabla_1^2 + \nabla_2^2) - e^2 \left(\frac{Z'}{r_1} + \frac{Z'}{r_2} \right) \right\rangle + \left\langle e^2 \left(\frac{Z'-2}{r_1} + \frac{Z'-2}{r_2} \right) \right\rangle + \left\langle \frac{e^2}{r_{12}} \right\rangle$$

$$= 2Z'^2 E_1 + e^2(Z'-2) \left\langle \frac{1}{r_1} \right\rangle + e^2(Z'-2) \left\langle \frac{1}{r_2} \right\rangle - \frac{5}{4} Z' E_1$$

$$\left[\left\langle \frac{1}{r_1} \right\rangle = \left\langle \frac{1}{r_2} \right\rangle = \frac{1}{\pi \left(\frac{a_0}{Z'}\right)^3} \int \frac{e^{-\frac{2Z'}{a_0}r}}{r} r^2 \sin\theta dr d\theta d\phi \right]$$

$$= 4 \left(\frac{Z'}{a_0}\right)^3 \int_0^\infty r e^{-\frac{2Z'}{a_0}r} dr$$

$$= -4 \left(\frac{Z'}{a_0}\right)^3 \frac{\partial}{\partial \alpha} \int_0^\infty e^{-\alpha r} dr, \quad \alpha = \frac{2Z'}{a_0}$$

$$= -4 \left(\frac{Z'}{a_0} \right)^3 \left[\frac{\partial}{\partial \alpha} \left(\frac{e^{-\alpha r}}{-\alpha} \right) \right] \Big|_0^\infty$$

$$= -4 \left(\frac{Z'}{a_0} \right)^3 \left[\frac{\alpha r e^{-\alpha r} + e^{-\alpha r}}{\alpha^2} \right] \Big|_0^\infty$$

$$= -4 \left(\frac{Z'}{a_0} \right)^3 \left(-\frac{1}{\alpha^2} \right)$$

$$= \cancel{4} \left(\frac{Z'}{a_0} \right)^3 \left(\frac{a_0}{Z'} \right)^2 \cdot \frac{1}{\cancel{4}}$$

$$= \frac{Z'}{a_0}$$

$$-4Z'^2 E_1 + 8Z' E_1$$

$$\Rightarrow \langle H \rangle = 2Z'^2 E_1 + \overbrace{2e^2(Z'-2) \cdot \frac{Z'}{a_0}}^{-4Z'^2 E_1 + 8Z' E_1} - \frac{5}{4} Z' E_1$$

$$= 2Z'^2 E_1 - 4Z'^2 E_1 + 8Z' E_1 - \frac{5}{4} Z' E_1$$

$$= \left[-2Z'^2 + \frac{27}{4} Z' \right] E_1$$

$$\Rightarrow E_{gs} = (-13.6 \text{ eV}) \left(-2Z'^2 + \frac{27}{4} Z' \right)$$

The parameter Z' is the effective charge of the nucleus as seen by one of the electrons, due to the presence of the other electron.

$$c) \quad \frac{dE_{gs}}{dz'} = (-13.6 \text{ eV}) \left(-4z' + \frac{27}{4} \right) = 0$$

$$\Rightarrow \boxed{z' = \frac{27}{16} = 1.6875}$$

$$\Rightarrow \boxed{E_{gs} = (-13.6 \text{ eV}) \left(-2 \cdot \frac{27^2}{16^2} + \frac{27^2}{64} \right)}$$

$$= (-13.6 \text{ eV}) \left(-\frac{27^2}{16 \cdot 8} + \frac{2 \cdot 27^2}{16 \cdot 8} \right)$$

$$= (-13.6 \text{ eV}) \left(\frac{3^6}{2^7} \right)$$

$$= \frac{(-13.6 \text{ eV})}{2} \left(\frac{3}{2} \right)^6$$

$$\Rightarrow \boxed{E_{gs} = -77.456 \text{ eV}}$$